

Playfair Cipher

3. Write a C program for Playfair algorithm is based on the use of a 5 X 5 matrix of letters constructed using a keyword. Plaintext is encrypted two letters at a time using this matrix.

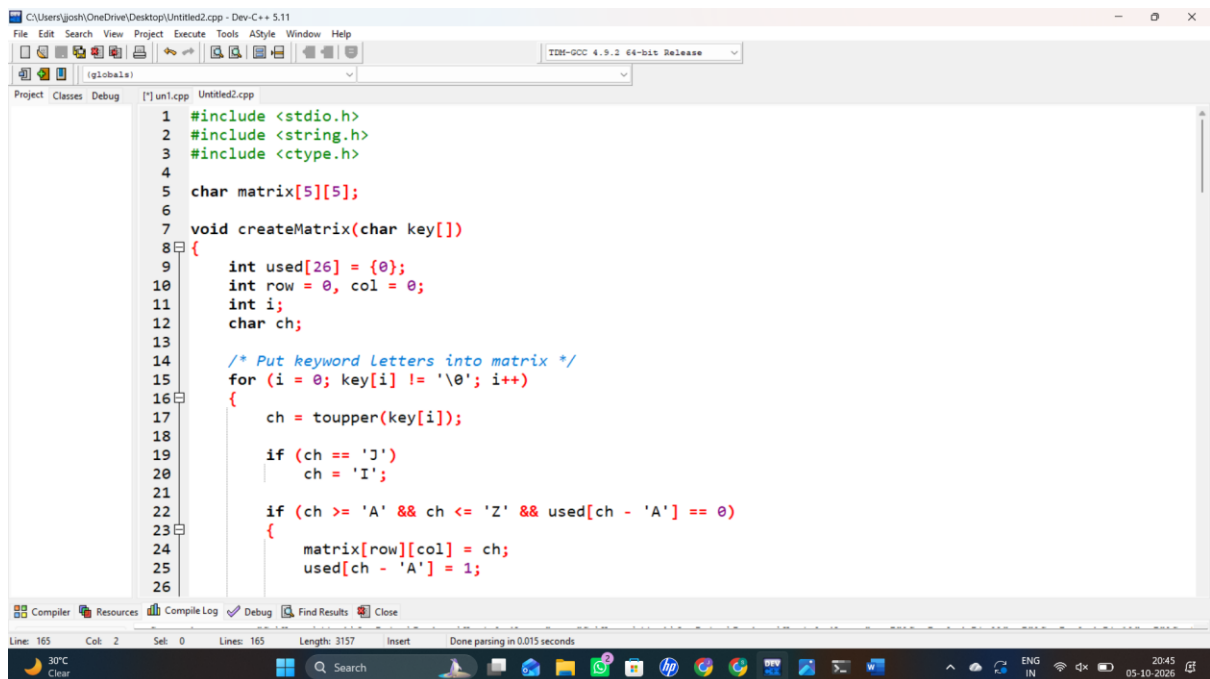
Aim

To write a C program to implement the **Playfair Cipher** using a **5 × 5 matrix** constructed from a keyword, and encrypt the plaintext two letters at a time.

Algorithm

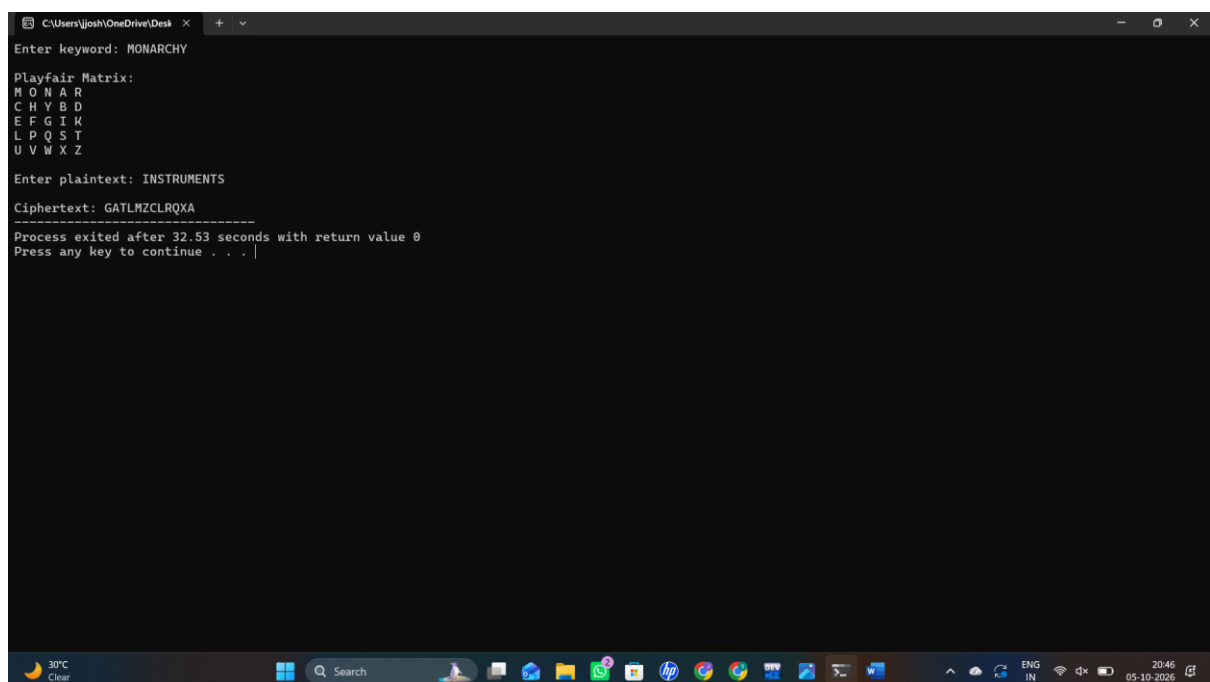
1. Start the program.
2. Read the keyword.
3. Construct a 5×5 matrix using the keyword.
4. Remove duplicate letters from the keyword.
5. Fill the remaining spaces with unused letters of the alphabet.
6. Combine I and J into one cell.
7. Divide the plaintext into pairs of letters.
8. If both letters in a pair are the same, insert X.
9. If the letters are in the same row, replace each with the letter to its right.
10. If the letters are in the same column, replace each with the letter below it.
11. Otherwise, form a rectangle and replace each letter with the letter in the same row but the other letter's column.
12. Display the ciphertext.
13. Stop.

Program



```
1 #include <stdio.h>
2 #include <string.h>
3 #include <ctype.h>
4
5 char matrix[5][5];
6
7 void createMatrix(char key[])
8 {
9     int used[26] = {0};
10    int row = 0, col = 0;
11    int i;
12    char ch;
13
14    /* Put keyword letters into matrix */
15    for (i = 0; key[i] != '\0'; i++)
16    {
17        ch = toupper(key[i]);
18
19        if (ch == 'J')
20            ch = 'I';
21
22        if (ch >= 'A' && ch <= 'Z' && used[ch - 'A'] == 0)
23        {
24            matrix[row][col] = ch;
25            used[ch - 'A'] = 1;
26        }
27    }
28 }
```

Input/Output



```
Enter keyword: MONARCHY
Playfair Matrix:
M O N A R
C H Y B D
E F G I K
L P Q S T
U V W X Z

Enter plaintext: INSTRUMENTS
Ciphertext: GATLMZCLRQXA
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Process exited after 32.53 seconds with return value 0
Press any key to continue . . .
```

Result

Thus, the **Playfair Cipher** was successfully implemented using a **5 × 5 matrix**, and the plaintext was encrypted two letters at a time.